ISO 9241-210 Human-Centred Design Report Template



MANIPAL INSTITUTE OF TECHNOLOGY
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Project Title

Course code and Course name

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Index

Contents

1. Introduction 3
1.1 Project Overview 3
1.2 Problem Statement 3
1.3 Objectives 3
1.4 Scope 4
2. Understanding the Context of Use 5
2.1 Target Users 5
2.2 User Environment 5
2.3 User Tasks 5
3. User Research 8
3.1 Personas 8
3.2 Empathy Map 8
4. User Requirements 10
4.1 Functional Requirements 10
4.2 Non-Functional Requirements 10
4.3 Prioritization 11
5. Design Solutions 12
5.1 Information Architecture 12
5.2 User Flow 14
5.3 Low-Fidelity Sketches 17
5.4 Wireframes 19
5.5 High-Fidelity Prototype 21
6. Human-Centred Design Principles Applied 23
6.1 Understand Users 23
6.2 User Involvement 23
6.3 Iterative Design 23
6.4 User Evaluation 24
6.5 Whole User Experience 24
7. Design Evaluation 25
7.1 Heuristic Evaluation 25
7.2 Usability Testing 25
7.3 Issues Identified 26
8. Design Improvements 28
9. Final Prototype 28
10. Conclusion 29
References 29

Page numbers are live fields and update automatically when the document is opened.

1. Introduction

1.1 Project Overview

CareerFlow is a student-centric campus placement and career management portal. Its tagline, “Your Career. Your Journey. Your Future.”, describes the four-step journey that organises the whole product: **Discover** drives, **Prepare** for them, **Apply** without mistakes, and **Succeed** by following each application through to an offer.

The product is a responsive web application for students. It is delivered as a fully interactive front end with realistic mock placement data, so that the design can be explored and evaluated without a back end. The work follows the human-centred design process of ISO 9241-210 and applies the HCI principles taught in this course.

Item	Summary
Product	CareerFlow, a campus placement and career management portal
Core journey	Discover → Prepare → Apply → Succeed
Platform	Responsive web application
Users	Students (placement officers and recruiters are out of scope)
Implementation	Interactive front end with mock data, no back end
Visual style	Modern SaaS dashboard: navy sidebar, blue actions, teal success, off-white pages, Inter typeface
Modules	Student Dashboard, Job Opportunities, Application Tracker, Eligibility Checker, Preparation Hub, Student Profile

1.2 Problem Statement

Campus recruitment asks students to manage many drives at once. Job details, eligibility rules, deadlines, preparation material and application status are usually spread across notice boards, emails, spreadsheets and chat groups.

This creates four recurring problems:

- **Unclear eligibility.** Students cannot quickly tell whether they meet a company’s CGPA, backlog, branch and score criteria, so they skip drives they qualify for or apply to drives they do not.
- **Application errors.** Choosing the wrong resume, missing a criterion or submitting by accident wastes a drive and often cannot be undone.
- **Poor visibility of status.** Students lose track of deadlines and of where each application stands.
- **High cognitive load.** Students must remember criteria and dates and move between several tools, instead of recognising them on screen.

Problem statement: how might we give students one place to discover drives, see at a glance whether they qualify, prepare, apply without errors and follow every application to its outcome?

1.3 Objectives

1. Reduce cognitive load by showing key job details and an eligibility badge on every job card (recognition rather than recall).
2. Minimise application errors with eligibility checks, a disabled Apply button that explains why, and a confirmation step.
3. Make system status visible through a five-stage tracker, preparation progress bars and a readiness score.
4. Provide a consistent, accessible interface with persistent navigation, standard status pills, keyboard support and sufficient contrast.
5. Cover the whole journey in six modules and demonstrate it as an interactive prototype.

Each objective has a measure based on the three usability attributes in ISO 9241-11. These are targets for the usability test in section 7.2, not results.

Objective	Attribute	Target
1	Efficiency	A student can judge their eligibility for a job from the job list in 10 seconds or less
2	Effectiveness	No applications are submitted to drives the student is not eligible for
3	Effectiveness	Students correctly state the current stage of an application
4	Satisfaction	System Usability Scale (SUS) score of 68 or higher
5	Effectiveness	All six modules are reachable and the main task in each can be completed

1.4 Scope

In scope	Out of scope
Six student-facing modules Responsive layout from phone to desktop Interactive front end with mock data for ten companies Keyboard operation, screen-reader labels and contrast checks	Placement officer and recruiter interfaces Back end, database and real authentication Email or push notifications Automatic resume parsing Native mobile apps

2. Understanding the Context of Use

2.1 Target Users

The primary users are undergraduate students in their final or pre-final year who are taking part in campus placements. They are comfortable with web and mobile apps, but they are under time pressure, are applying to several companies at once and may be anxious about missing an opportunity.

User group	Role in the system	Designed for in this project?
Students (final and pre-final year)	Discover drives, check eligibility, prepare, apply and track applications	Yes, primary users
Placement officers	Publish drives, set criteria, review applications	No, future work
Recruiters	Review shortlisted candidates and schedule rounds	No, future work

Characteristics that shape the design:

- **Mixed branches and goals.** Software, analyst and core engineering students need different criteria and different preparation.
- **Mixed devices.** Students switch between a laptop in the hostel or library and a phone between classes.
- **Accessibility needs.** Some students have low vision, colour-blindness or rely on a keyboard or screen reader, so the interface follows universal design principles.
- **Limited patience for errors.** A wrong or missed application can cost an opportunity, so the interface must prevent and explain errors.

2.2 User Environment

Context factor	Description	Design implication
Physical	Hostel rooms, libraries, labs and corridors; frequent interruptions	Short, scannable screens; progress is easy to resume
Technical	Laptops and mid-range phones, variable campus Wi-Fi, modern browsers	Responsive layout, lightweight pages, instant client-side feedback
Time	Deadlines cluster in placement season	Deadlines visible on the dashboard and on every job card
Organisational	Companies set criteria; the placement cell sets rules and dates	Criteria shown exactly as published; eligibility checked before applying
Social	Peers compare offers; pressure and anxiety are common	Calm tone, positive feedback, clear next steps

2.3 User Tasks

User tasks were analysed with hierarchical task analysis (HTA). The top-level goal, “get placed through a campus drive”, is split into five tasks. Each task has a plan that states the order in which its subtasks happen, and each maps to one CareerFlow module.

Task	Plan	Subtasks	Module
1. Discover drives	1.1, then 1.2 or 1.3	Search, filter or sort; read the eligibility badge; open company details; save a job	Job Opportunities
2. Check eligibility	2.1 to 2.3, then 2.4 if needed	Pick a company and role; confirm academic details; read the pass or fail table; note missing skills	Eligibility Checker
3. Prepare	Any order, repeated	Solve DSA problems; take an aptitude test; tick the resume checklist; book a mock interview	Preparation Hub
4. Apply	4.1 to 4.4 in order	Press Apply Now; choose a resume; tick the confirmation; submit and read the toast	Job Opportunities
5. Track	5.1, 5.2, then 5.3 or 5.4	Open the tracker; read stage and next step; act on a deadline alert; accept an offer	Application Tracker

Plan 0 for the whole goal is: repeat tasks 1 and 2 until a drive fits, do task 3 alongside, then do task 4, then repeat task 5 until there is a result. **Task 4 (Apply) carries the highest risk of error**, because it is hard to undo. The design answers this with a disabled Apply button that explains why and a confirmation dialog.

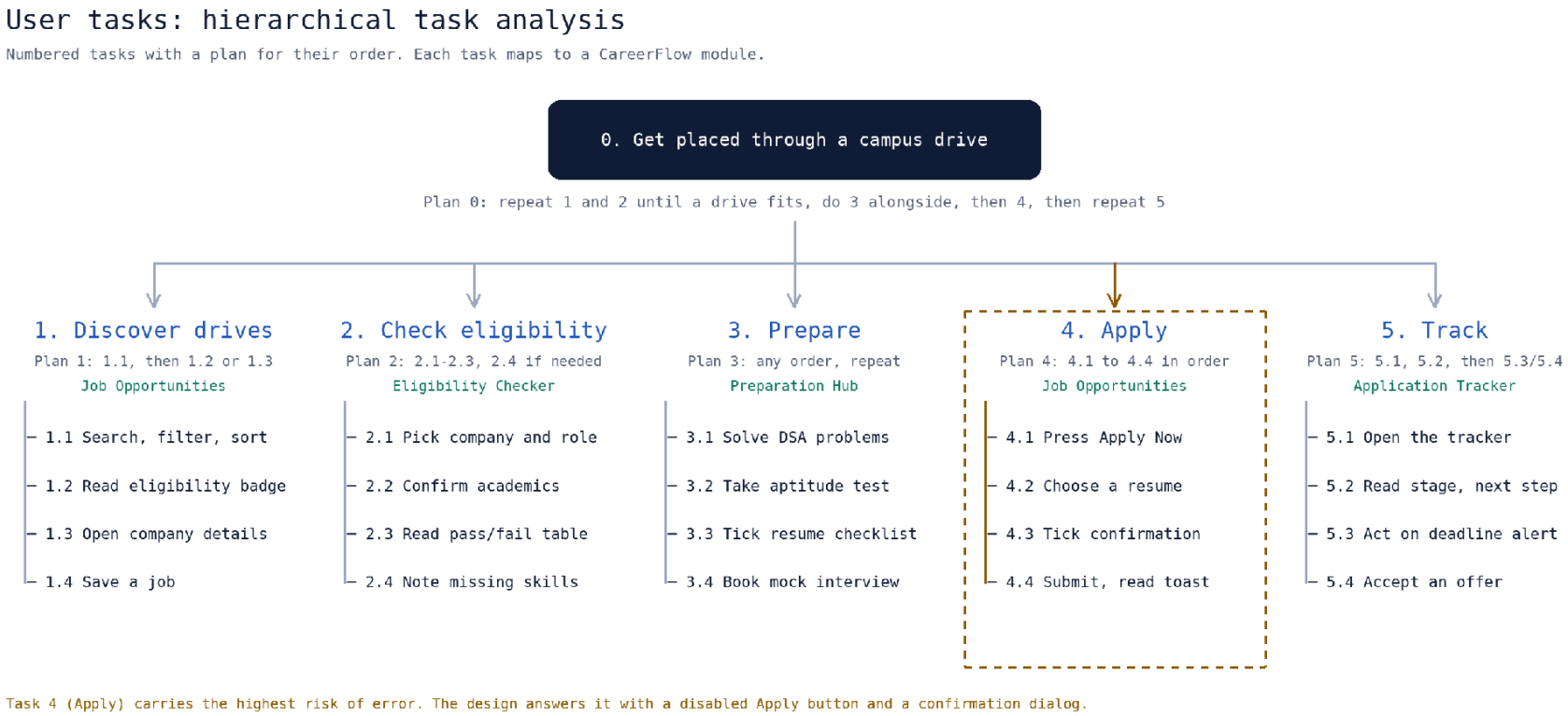


Figure 1: Hierarchical task analysis for getting placed through a campus drive. The dashed box marks the highest-risk task.

3. User Research

3.1 Personas

The three personas below are proto-personas. They are built from the context of use in section 2 and the course material, not from interviews, so they should be validated with real students before they are treated as evidence.

Persona 1: Aarav Mehta, 21, Computer Science (software focus)

Attribute	Description
Goal	Land a software engineer offer at a product company
Frustrations	Is never sure which drives he qualifies for; misses deadlines because dates are in emails and chats
Behaviour	Uses a laptop for preparation and his phone for quick checks; practises DSA daily
Needs	Eligibility at a glance, a clear deadline list, a DSA tracker, readiness feedback
Design response	Eligibility badge on job cards, deadlines widget, DSA tracker and readiness score

Persona 2: Meera Nair, 21, Electronics and Communication (analyst roles)

Attribute	Description
Goal	Move into a business or data analyst role although her branch is not computer science
Frustrations	Branch rules differ by company; unsure which skills she lacks
Behaviour	Compares several roles before applying; polishes her resume repeatedly
Needs	Branch-aware eligibility, a skills match with named gaps, a resume checklist
Design response	Eligibility Checker with skills gaps, resume checklist, role-type filter

Persona 3: Rohan Kulkarni, 22, Mechanical Engineering (core roles, mobile first)

Attribute	Description
Goal	Secure a core engineering position and understand how his one backlog affects eligibility
Frustrations	Portals that are hard to use on a phone; unclear backlog rules
Behaviour	Does most tasks on his phone with unreliable connectivity
Needs	Responsive layout, explicit backlog criteria, simple navigation
Design response	Mobile layout with scrolling navigation, criteria shown as numbers (for example “0 backlogs max”)

3.2 Empathy Map

The empathy map below is for Persona 1 (Aarav) during placement season.

Says	Thinks
"Am I even eligible for this?" "When does this close?" "Which round am I in now?"	Missing one deadline could cost me the whole season Others seem to be ahead of me I need one place to see everything
Does	Feels
Scrolls through emails and chat groups for dates Keeps a personal spreadsheet Practises DSA and mock tests at night	Anxious about missing drives Confused by scattered information Relieved when a status is clear
Pains	Gains
Unclear criteria, duplicate effort, wrong or missed applications, no visible progress	Instant eligibility answers, one deadline list, visible progress, confidence that an application was received

4. User Requirements

4.1 Functional Requirements

ID	Requirement	Module
FR1	Personalised welcome banner with the Discover, Prepare, Apply, Succeed stepper	Dashboard
FR2	Counts of applied, shortlisted and offered applications	Dashboard
FR3	Upcoming deadlines widget with days left	Dashboard
FR4	Notifications list with unread markers and mark all read	Dashboard
FR5	Recommended jobs the student is eligible for	Dashboard
FR6	Search by company, role or skill	Jobs
FR7	Filter by role type, minimum CTC, location and eligibility; sort by deadline or CTC	Jobs
FR8	Save and unsave jobs	Jobs
FR9	Job cards show CTC, location, deadline and an eligibility badge	Jobs
FR10	Company details with criteria, pass or fail rows and skills match	Jobs
FR11	Apply with a confirmation step; Apply is disabled with a reason when not eligible	Jobs
FR12	Five-stage application board with progress bars	Tracker
FR13	Next-step message for each application	Tracker
FR14	Choose a company and role to check	Eligibility
FR15	Compare CGPA, backlogs, branch, 10th and 12th scores with the criteria	Eligibility
FR16	Pass or fail breakdown with a count of criteria met	Eligibility
FR17	Skills match with named gaps	Eligibility
FR18	Interview resources that can be marked done	Preparation
FR19	DSA practice tracker by topic	Preparation
FR20	Aptitude scores and practice	Preparation
FR21	Interactive resume checklist	Preparation
FR22	Book and cancel a mock interview slot	Preparation
FR23	Edit personal and academic details with validation	Profile
FR24	Add and remove skill tags	Profile
FR25	Upload (PDF, up to 2 MB), preview and delete resumes	Profile
FR26	Choose job preferences	Profile

4.2 Non-Functional Requirements

ID	Quality	Requirement
NFR1	Usability	Key job details and eligibility are visible without opening anything
NFR2	Accessibility	Text contrast of at least 4.5:1, full keyboard operation, visible focus, ARIA labels, and status never shown by colour alone
NFR3	Responsiveness	Works from phone width to wide desktop screens
NFR4	Performance	Interactions respond instantly because the prototype runs entirely in the browser
NFR5	Consistency	Reusable cards, status pills, buttons and navigation across all modules
NFR6	Feedback	Every save, apply, upload, booking or error produces a visible message
NFR7	Error handling	Validation messages say what is wrong and how to fix it
NFR8	Compatibility	Current versions of Chrome, Edge, Firefox and Safari
NFR9	Privacy	Only mock data is used; a real release needs secure storage of student data

4.3 Prioritization

Requirements are prioritised with the MoSCoW method. Priority follows the core journey and the need to prevent application errors.

Priority	Requirements	Reason
Must have	FR6, FR7, FR9, FR10, FR11, FR12, FR15, FR16, FR23, FR25; NFR1, NFR2, NFR6	Without these the journey, eligibility checks and error prevention do not work
Should have	FR1, FR3, FR4, FR8, FR13, FR14, FR17, FR19, FR21, FR24; NFR3, NFR5, NFR7	They reduce effort and increase confidence but the journey still works without them
Could have	FR2, FR5, FR18, FR20, FR22, FR26; NFR4, NFR8	Useful extras that support preparation and personalisation
Won’t have (this release)	Officer and recruiter views, back end, real notifications, resume parsing; NFR9 beyond mock data	Outside the student-only scope

5. Design Solutions

5.1 Information Architecture

The structure is deliberately shallow: one persistent shell with six top-level modules. Six items stay within what a student can scan at once, and every module is one click from anywhere. A top header on every screen holds global search, notifications and a profile shortcut.

Module	Journey stage	Purpose and content
Dashboard	Overview	Welcome and stepper, counts, deadlines, notifications, recommended jobs
Job Opportunities	Discover	Search, filters, sort, job cards with eligibility badge, company details, apply confirmation
Application Tracker	Apply and succeed	Five stages: Applied, Shortlisted, Assessment, Interview, Selected
Eligibility Checker	Discover	Company picker, academic inputs, skills match, pass or fail breakdown
Preparation Hub	Prepare	Interview resources, DSA tracker, aptitude, resume checklist, mock interviews
Student Profile	Your details	Personal and academic details, skill tags, resumes, preferences

Cross-links connect the modules the way students think: dashboard counts open the Tracker, a job card opens the Eligibility Checker, profile academics feed the checker, and a submitted application lands in the Tracker. Labels use words students already know (Jobs, Tracker, Eligibility, Prep).

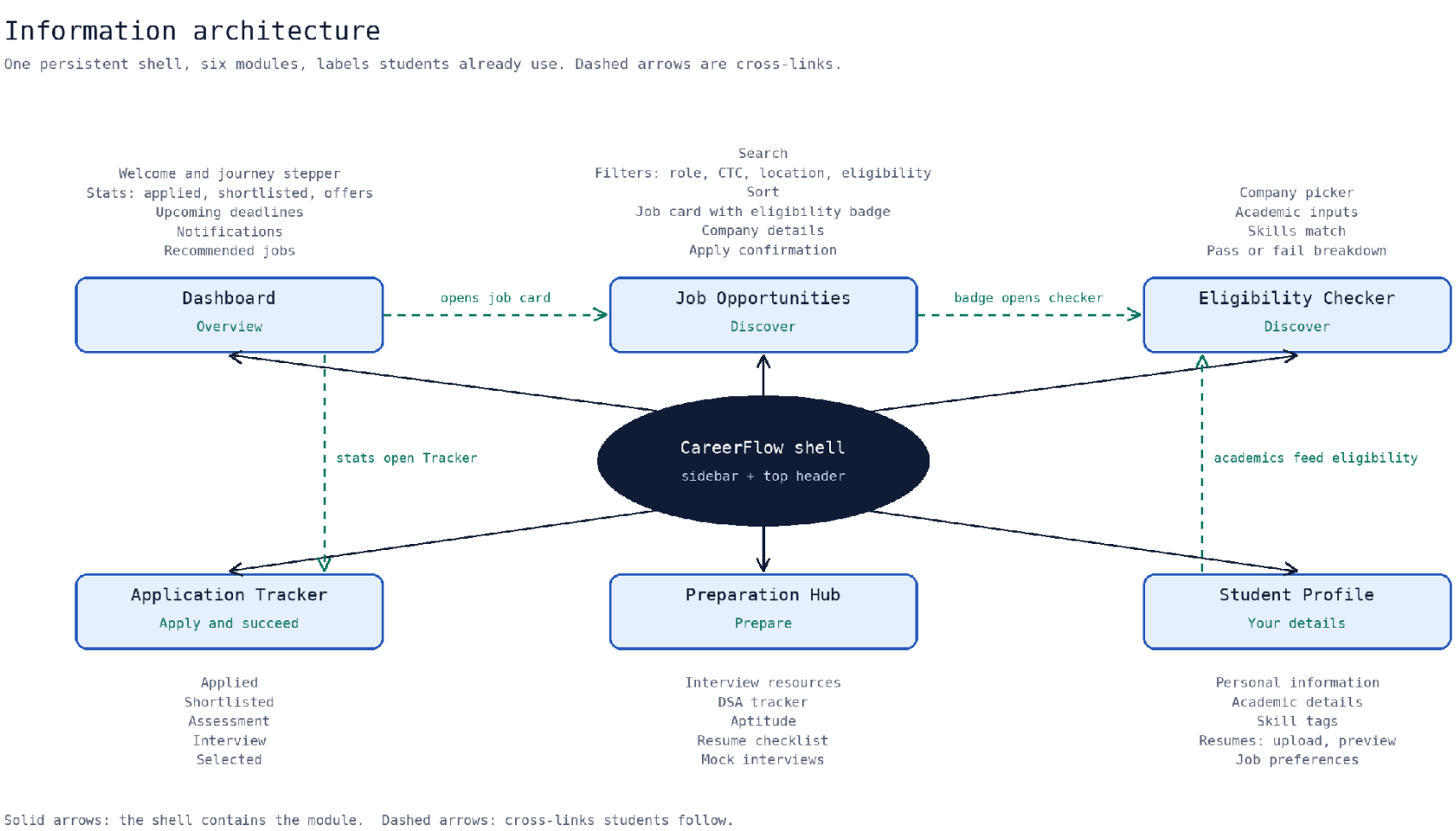


Figure 2: Information architecture. Solid arrows show the shell containing each module; dashed arrows show cross-links.

5.2 User Flow

Four flows cover the journey, shown in Figure 3:

- **Discover:** open the app, reach Job Opportunities, search and filter, open a job card, and meet the first decision: is the student eligible?
- **Apply:** press Apply Now, review the confirmation dialog, tick the confirmation and submit. An unticked box produces an error toast and the dialog stays open. A successful submission shows a toast and the Tracker shows Applied.
- **Recover:** if the student is not eligible, Apply is disabled and the reason is shown. The Eligibility Checker shows what is missing; the student then updates the profile and returns to Jobs, or picks another job.
- **Prepare:** the dashboard shows a readiness score, which leads to the Preparation Hub. Completing an item updates the progress bar and the score.

Dialogue states (state transition network)

The state transition network in Figure 4 defines the states and events of three interactions: the stage of an application, the Apply control, and saving a job. It makes two design rules from the course explicit. **Observability:** every state is visible to the student (a stage badge, a disabled button with a reason, a filled star). **Recoverability:** the student can leave an unwanted state, for example by cancelling the dialog with Cancel or Esc, or by pressing the star again to remove a saved job.

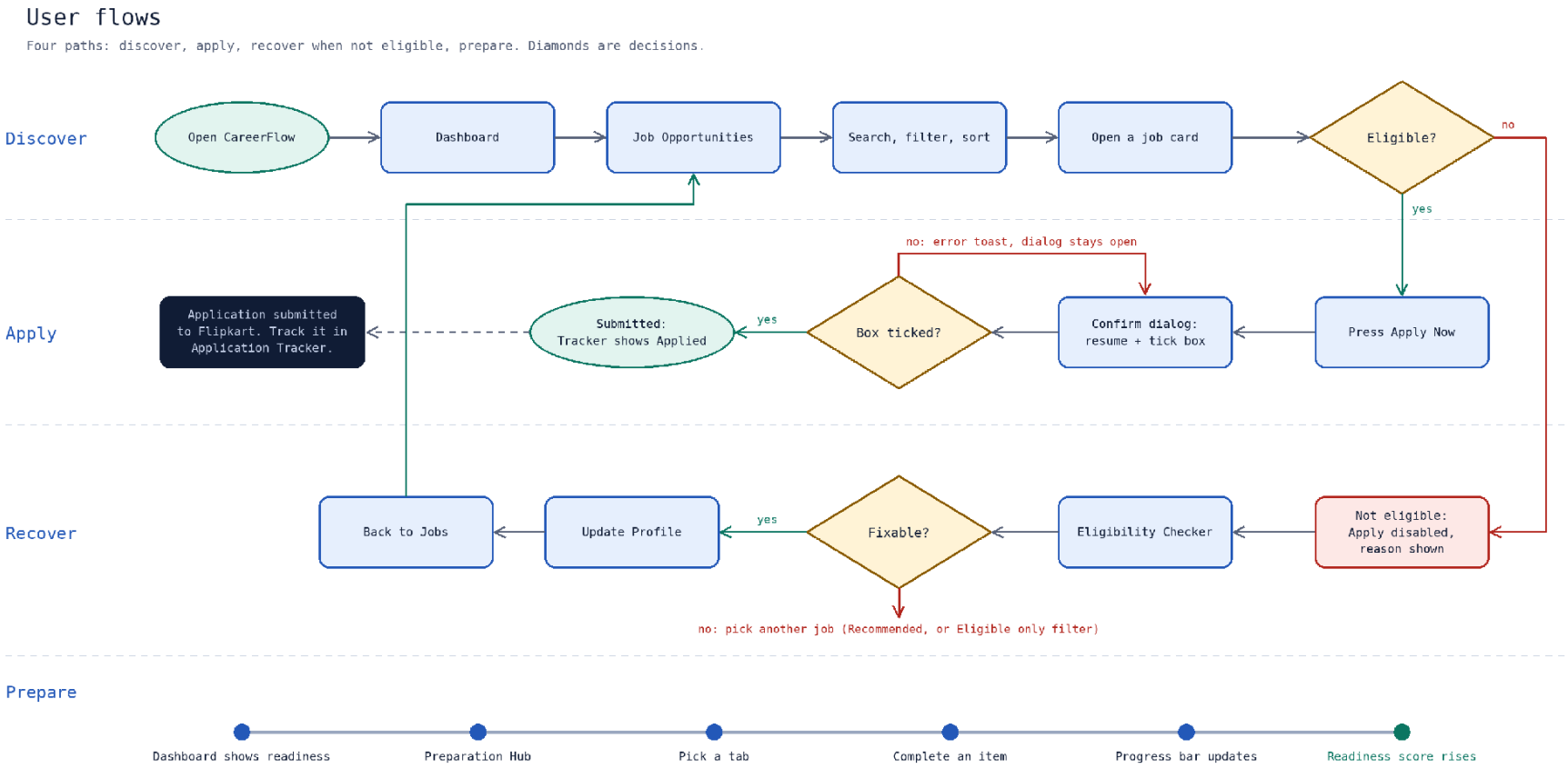


Figure 3: User flows for discovering, applying, recovering when not eligible, and preparing.

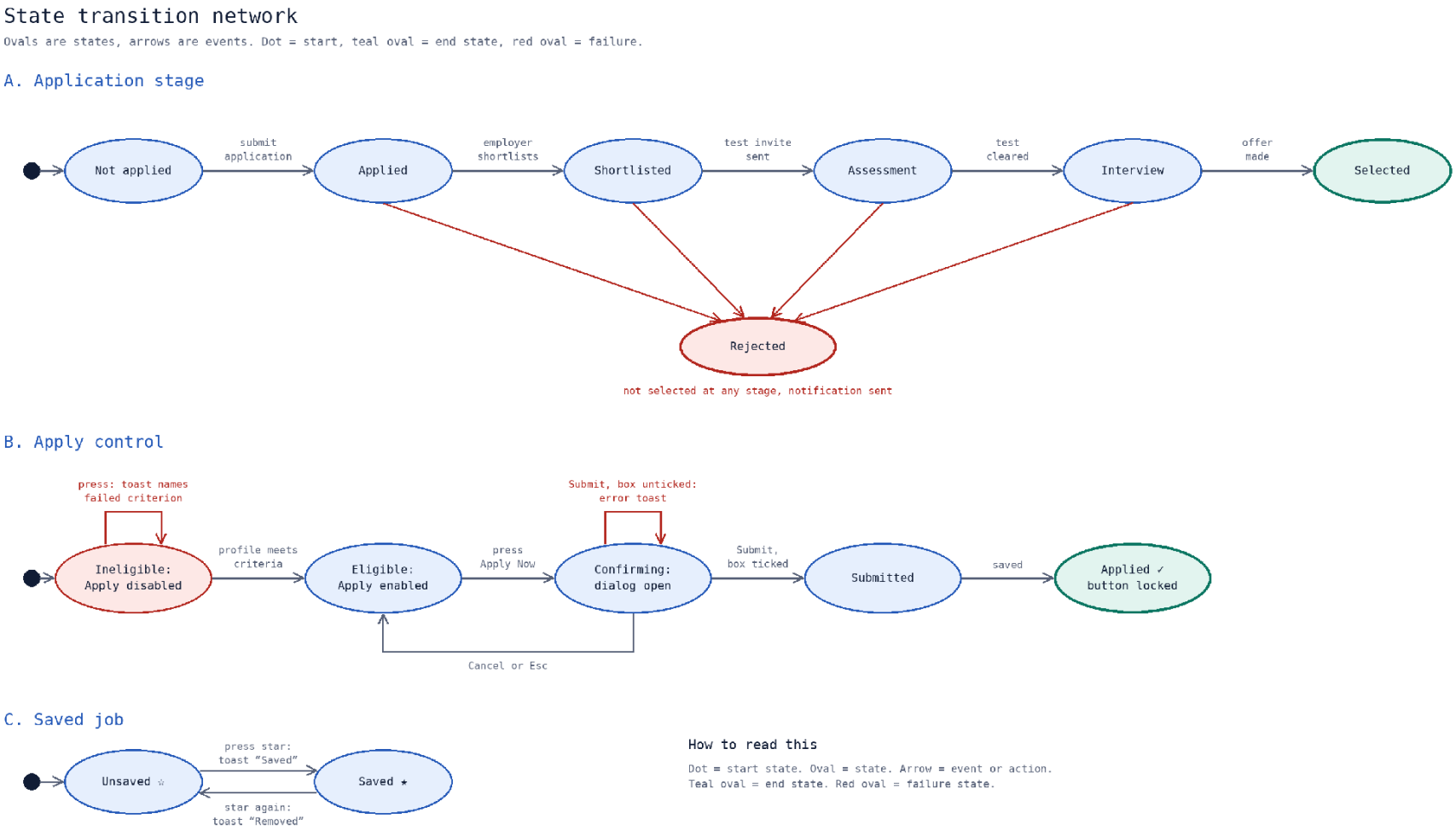


Figure 4: State transition network for application stage, the Apply control and saved jobs.

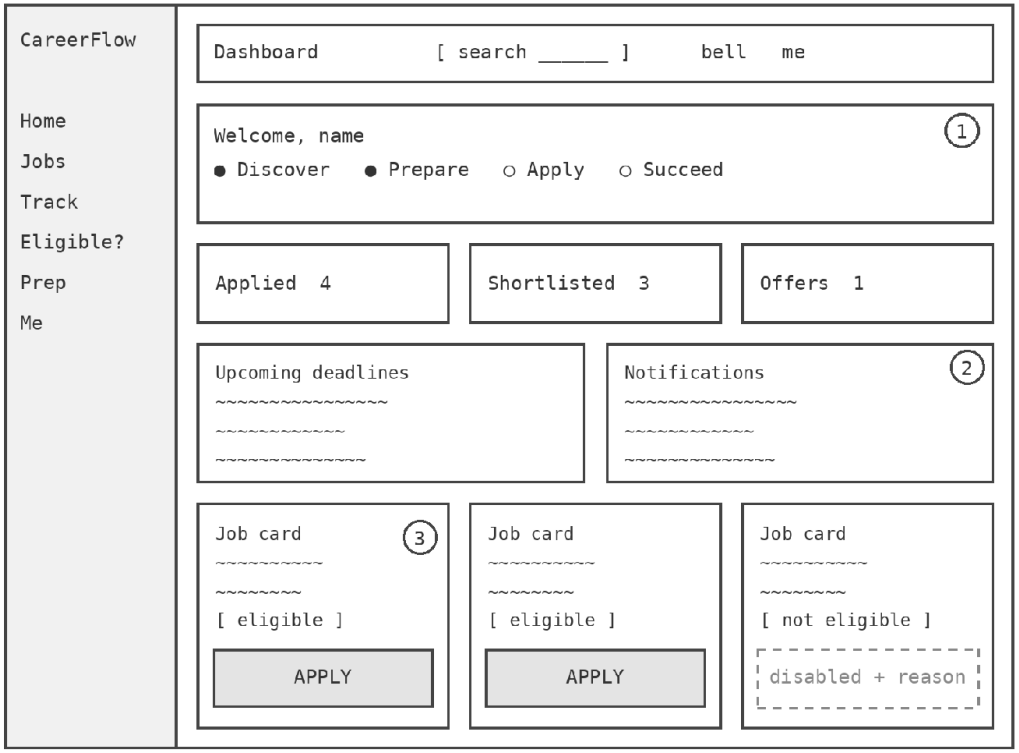
5.3 Low-Fidelity Sketches

Low-fidelity sketches were used to settle the layout of the Dashboard on desktop and phone before any detail was drawn. They fixed four decisions: a left sidebar on desktop that becomes a scrolling bar on phones; a journey stepper at the top of the dashboard; related information grouped into cards; and a large Apply button that greys out with a reason.

Two design questions were recorded with the Question, Options, Criteria (QOC) notation from the course.

Question	Options	Recognition	Consistency	Small screens	Decision
How do students move between six modules?	Persistent left sidebar	Good: labels always visible	Good	Collapses to a scrolling bar	Chosen
	Top tabs	Good	Good	Six long labels do not fit	
	Hamburger menu	Poor: labels hidden	Good	Good	
How is application progress shown?	Five-stage board	Good: all stages visible	Good	Scrolls sideways on phones	Chosen
	List with status text	Fair	Good	Good	
	Stepper per application	Good	Fair	Takes much space	

Low-fidelity sketches: Dashboard





Mobile: one column, nav scrolls sideways

1 stepper shows where I am (visibility of status)

2 cards keep related things together (grouping)

3 APPLY is big, and greys out with a reason (error prevention)

Figure 5: Low-fidelity sketches of the Dashboard on desktop and mobile, with numbered design notes.

5.4 Wireframes

The wireframe of Module 01 (Dashboard) uses real labels and spacing but no colour. Numbered markers tie each part of the screen to the HCI principle it supports.

No.	Principle	Where it appears on the screen
1	Visibility of system status	Journey stepper, counts and the readiness percentage
2	Recognition rather than recall	Job cards show CTC, deadline, criteria and the eligibility badge
3	Fitts's law	Apply Now is 48 px tall and takes half of the card width
4	Gestalt principles	Related data is grouped in cards with equal 16 px gaps
5	Error prevention	Ineligible jobs show a disabled Apply with the reason
6	Consistency	Same sidebar, pills and cards in all six modules
7	Universal design	4.5:1 text contrast, visible focus ring, labelled controls
8	Feedback	A toast confirms saves, applications, uploads and errors

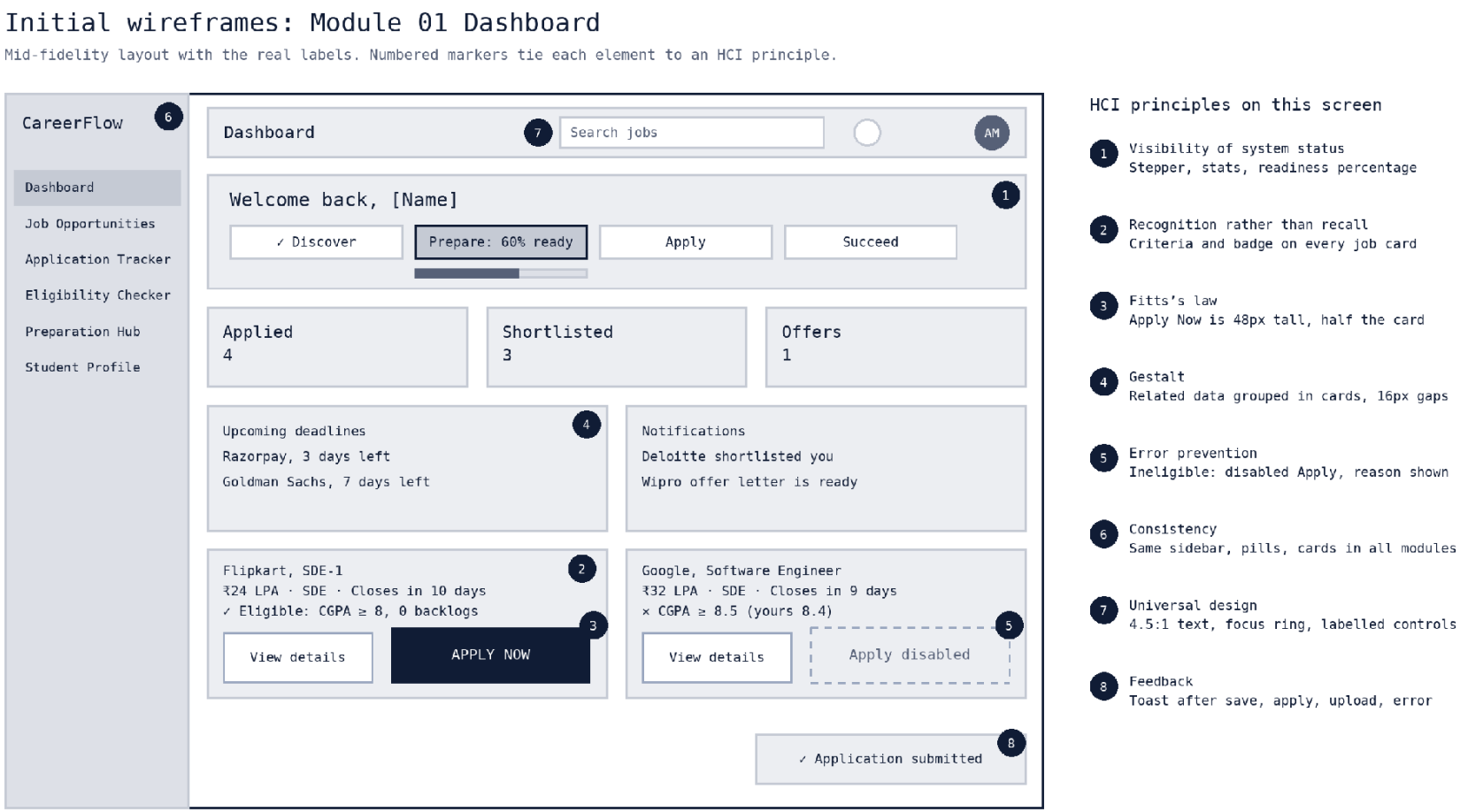


Figure 6: Initial wireframe of Module 01 (Dashboard) with HCI principle markers.

5.5 High-Fidelity Prototype

The high-fidelity design applies the visual identity: navy (#101C35) for the sidebar, blue (#3578E5) for primary actions, off-white (#F4F6FA) for pages, and teal (#16A085) for success. Typography is the Inter sans-serif family on a generous spacing scale, with soft borders and shadows on cards. The colour pairs below were checked against the WCAG 2.1 AA text contrast minimum.

Colours (text on background)	Ratio	Where it is used	Text AA (4.5:1)
#101C35 on #F4F6FA	15.7:1	Body text on page background	Pass
#566077 on #F4F6FA	5.8:1	Secondary text on page background	Pass
#FFFFFF on #3578E5	4.2:1	Apply Now and Submit button text	Below 4.5:1
#FFFFFF on #0B7662	5.6:1	Completed journey step	Pass
#0B7662 on #E2F4EF	4.9:1	Eligible badge	Pass
#8A5A00 on #FFF3D6	5.4:1	Deadline warning pill	Pass
#B3261E on #FDE8E6	5.6:1	Not eligible badge	Pass
#2358B8 on #E6EFFD	5.7:1	Information pill and link text	Pass
#C9D4EC on #101C35	11.4:1	Sidebar labels	Pass
#16A085 on #FFFFFF	3.3:1	Teal on white (used for fills and bars, never for text)	Not used for text

Teal is used for fills and progress bars only. Wherever teal carries text it is darkened to #0B7662 so it passes. One pair falls short: white text on the specified blue (#3578E5) reaches 4.2:1. This is recorded as issue I3 in section 7.3.

The interactive prototype contains all six modules:

Module	What can be done in the prototype
Dashboard	Read the stepper, counts, deadlines and notifications; mark notifications read; open recommended jobs
Job Opportunities	Search, filter, sort, save jobs, open details and apply through a confirmation dialog
Application Tracker	See applications by stage with progress bars; advance a stage in demo mode
Eligibility Checker	Change CGPA, backlogs, branch and scores and see the pass or fail table update
Preparation Hub	Mark resources done, log DSA problems, take practice tests, tick the checklist, book a mock interview
Student Profile	Edit details with validation, add or remove skills, upload, preview and delete PDF resumes, set preferences

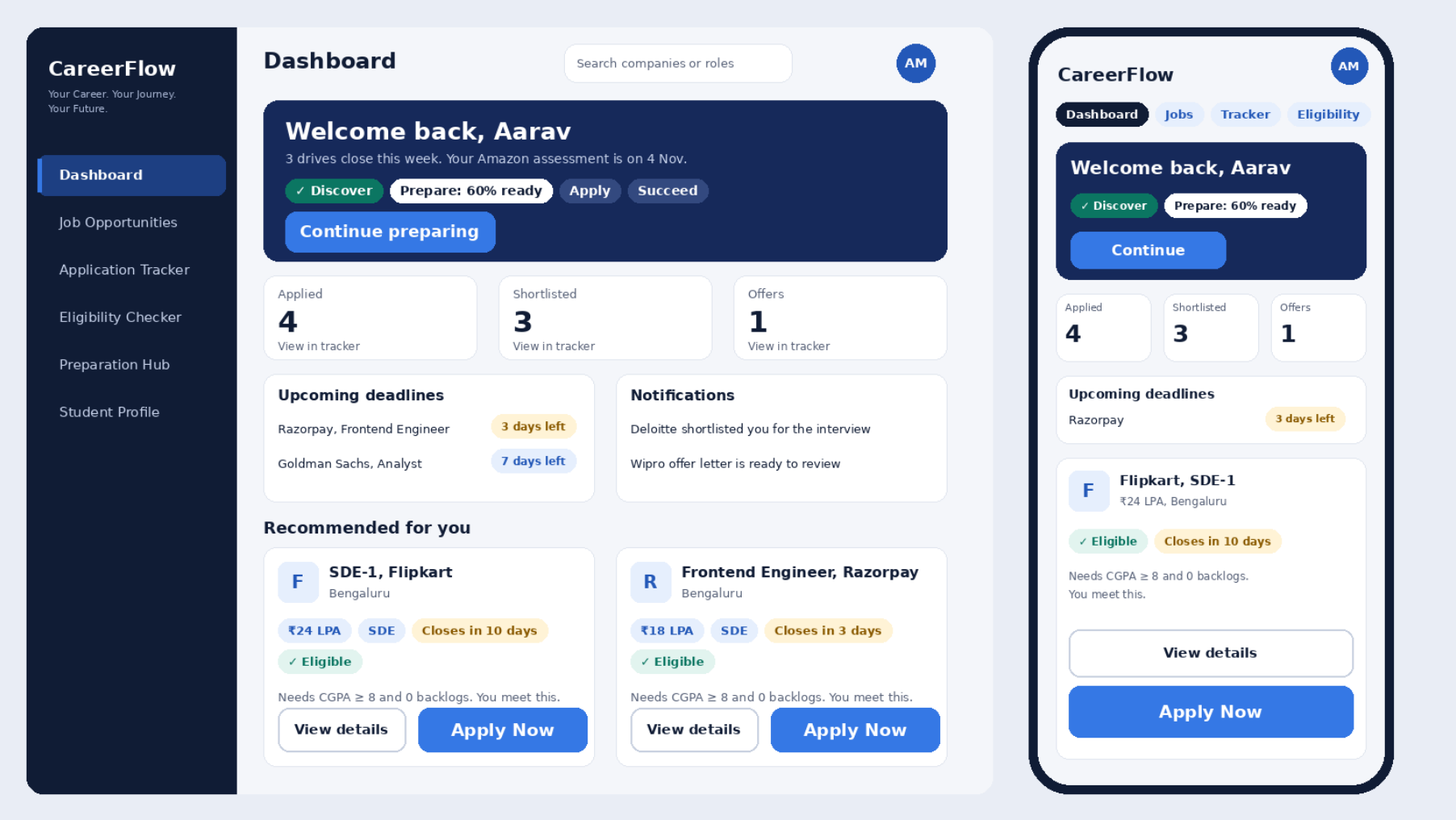


Figure 7: Final screens for Module 01 (Dashboard) on desktop and mobile.

6. Human-Centred Design Principles Applied

ISO 9241-210 describes human-centred design as an iterative cycle: understand and specify the context of use, specify the user requirements, produce design solutions, and evaluate them against the requirements. The cycle maps onto this report as follows.

ISO 9241-210 activity	Section of this report
Understand and specify the context of use	2 and 3
Specify the user requirements	4
Produce design solutions	5
Evaluate the designs against requirements	7, with changes in 8 and the result in 9

6.1 Understand Users

The design starts from an explicit description of users, tasks and environment: the context-of-use tables, the task analysis, three personas and an empathy map. These drove concrete decisions, for example showing eligibility on the job card because students struggle to judge it, and giving Task 4 extra protection because it is the riskiest. The limitation is that the personas rest on assumptions and should be validated with real students.

6.2 User Involvement

In this iteration users are represented by the proto-personas, and no student has yet been involved directly.

Planned involvement:

- Interview five to six students from different branches to validate the personas and problem statement.
- Run a card sort to confirm the module labels.
- Ask a placement officer to check that the eligibility criteria match real drives.
- Include students in the usability test described in section 7.2.

6.3 Iterative Design

The design moved through increasing fidelity. Each step used the one before it and exposed problems early and cheaply.

Step	Artefact	What it settled
1	Context of use and HTA	Which tasks matter and which is riskiest
2	Information architecture	Six modules, one persistent shell, cross-links
3	User flows	The decisions that need protection: eligibility and confirmation

Step	Artefact	What it settled
4	State transition network	Every state and its way back, including error states
5	Sketches and wireframes	Layout, hierarchy and where each HCI principle appears
6	Interactive prototype	Behaviour that can be inspected and evaluated

The next iteration is driven by the evaluation in section 7 and is set out in section 8.

6.4 User Evaluation

The design has so far been evaluated by an expert using Nielsen's heuristics (section 7.1). Evaluation with real users is planned in section 7.2 and will use the ISO 9241-11 measures of effectiveness, efficiency and satisfaction. Both feed the improvements in section 8.

6.5 Whole User Experience

The design covers the whole experience rather than single screens. The journey stepper keeps the end goal in view; feedback messages are calm and specific; the interface adapts to phones; and accessibility is built in through contrast, keyboard operation and non-colour cues. The multidisciplinary principle of ISO 9241-210 is acknowledged in the design workflow, which combines task analysis, interaction design, visual design and front-end development.

7. Design Evaluation

7.1 Heuristic Evaluation

One evaluator, the author, walked through the four core tasks (find a job, check eligibility, apply, track an application) in the prototype and judged each of Nielsen's ten heuristics. Severity uses the course scale of 0 (not a problem) to 4 (usability catastrophe). Severity ratings are the evaluator's estimate; a single evaluator finds only some of the problems, so three to five evaluators are recommended.

Heuristic	Evidence in the design	Finding
1. Visibility of system status	Journey stepper, five-stage tracker, progress bars, toasts	Good
2. Match with the real world	Placement vocabulary: drives, CTC, backlogs, shortlisted	Terms such as CTC and DSA are not explained (I8)
3. User control and freedom	Esc and Cancel close dialogs; bookings can be cancelled	Submitted applications cannot be withdrawn, which is by design; saved jobs cannot be reviewed (I1)
4. Consistency and standards	Same sidebar, pills and cards in every module	Tabs do not follow standard arrow-key behaviour (I4)
5. Error prevention	Disabled Apply with reason, confirmation dialog, input validation	Checker verdict and Apply button can disagree (I5)
6. Recognition rather than recall	Criteria and badge on each job card	Good
7. Flexibility and efficiency	Header search, filters, sort, URL per module	Search is hidden on narrow screens (I6)
8. Aesthetic and minimalist design	Card layout, generous spacing, one accent colour	Good
9. Help users recover from errors	Specific messages that name the failed criterion	Error toasts vanish after about four seconds (I7)
10. Help and documentation	None	No onboarding or glossary (I8)

7.2 Usability Testing

No user testing has been carried out yet. This section is the test plan, with a results table to complete after the sessions.

Method: moderated, task-based sessions with think-aloud, about 20 minutes each, on a laptop and a phone. **Participants:** five to eight final-year students from at least three branches, including one who mainly uses a phone and, if possible, one who uses a keyboard or screen reader. **Measures:** effectiveness (task completion), efficiency (time on task and errors) and satisfaction (SUS questionnaire).

ID	Task	Success criterion	Target time
T1	Find the jobs paying at least ₹10 LPA that you are eligible for	Names at least three eligible jobs	60 s
T2	Find out why you cannot apply to Google and what would change that	States the failed criterion	60 s
T3	Apply to Flipkart SDE-1	Application submitted and visible in the Tracker	90 s
T4	Say what stage the Deloitte application is at	Correct stage and next step	30 s
T5	Book a mock interview slot	Booking confirmed	45 s
T6	Upload a new resume and update your CGPA	Both saved with confirmation	90 s

Illustrative dummy results (replace with actual session data):

The following values are dummy test cases for demonstrating how the results section can be reported; they are not observations from real participants.

Task	Completed (n of N)	Mean time	Errors	Notes
T1	6 of 6	42 s	1	One participant initially missed the eligibility filter; all named at least three eligible jobs after the prompt.
T2	5 of 6	51 s	2	Two participants checked the profile before opening the eligibility explanation.
T3	6 of 6	78 s	1	One participant forgot the confirmation checkbox on the first attempt; no accidental submission occurred.
T4	6 of 6	19 s	0	All participants stated the current stage and next step correctly.
T5	5 of 6	39 s	1	One participant needed a reminder to open the Preparation Hub before booking.
T6	5 of 6	83 s	2	Two participants needed help locating the resume upload control on mobile width.
SUS score	—	—	—	74 / 100 average (illustrative; above the target of 68).

7.3 Issues Identified

These issues come from the heuristic evaluation and a review of the prototype. They are ordered by severity.

ID	Issue	Module	Heuristic	Severity
I1	Jobs can be saved but there is no way to view saved jobs later	Jobs, Dashboard	Flexibility; user control	3
I2	Focus is not trapped in dialogs, so keyboard users can tab to the page behind	All dialogs	Universal design; error prevention	3
I3	Primary button text (white on #3578E5, 15 px bold) has 4.2:1 contrast, below the 4.5:1 minimum for normal text	Apply, Submit	Universal design	2
I4	Preparation tabs do not respond to arrow keys	Preparation	Consistency and standards	2
I5	Checker can say “Eligible” for what-if values while Apply follows the saved profile, without explanation	Eligibility	Consistency; error recovery	2
I6	Header search is hidden on narrow screens	Shell	Flexibility and efficiency	2
I7	Toasts vanish after about four seconds, including error messages	All	Error recovery	2
I8	No help or glossary for terms such as CTC, DSA and LPA	All	Help and documentation	2
I9	Data is not kept between visits (prototype limitation)	All	User control	1
I10	A “Demo: advance stage” button exposes prototype logic	Tracker	Match with the real world	1

8. Design Improvements

Each issue has a planned improvement. These changes have not yet been built into the prototype that was evaluated, so they are listed as planned.

Issue	Improvement	Principle	Status
I1	Add a “Saved only” filter and a saved-jobs count on the Dashboard	Flexibility and efficiency	Planned
I2	Trap focus inside open dialogs and return focus to the trigger on close	Universal design	Planned
I3	Raise Apply Now and Submit to large bold text (at least 18.7 px) so 3:1 applies, or darken the button text background to #2F6FDB, which reaches 4.8:1	Universal design	Planned
I4	Add arrow, Home and End key support to tabs with a single tab stop	Consistency and standards	Planned
I5	Label the checker as “what-if”, and when the saved profile fails show “Update Profile to apply” instead of Apply	Error prevention	Planned
I6	Show a search icon on narrow screens that expands on tap	Flexibility and efficiency	Planned
I7	Keep error messages on screen until dismissed and add a close button	Error recovery	Planned
I8	Add short tooltips for CTC, DSA and LPA and a first-run “How CareerFlow works” card	Help and documentation	Planned
I9	Keep prototype state between visits using local storage	User control	Planned
I10	Move demo controls behind a “Demo mode” switch	Match with the real world	Planned

After these changes the test plan in section 7.2 should be run and the results compared with the targets in section 1.3.

9. Final Prototype

The current prototype is a single-page web application with all six modules and mock data for ten companies. It is version 1: it includes the design decisions in sections 5 and 6 but not yet the improvements in section 8. Figure 7 shows the final Dashboard screens.

Interactive prototype: https://claude.ai/artifact/Bp41qoitktBpsmAVREUKdA

Suggested walkthrough:

1. Open **Job Opportunities** and tick “Eligible jobs only”. Google and Goldman Sachs drop out because the student’s CGPA is 8.4 and they need 8.5.
2. Clear the filter and press the disabled **Apply** on Google. A message names the failed criterion.

3. Open the **Eligibility Checker**, choose Google and change the CGPA to 8.6. The table turns to pass and Apply becomes possible in the what-if view.
4. Go back to Jobs and apply to Flipkart. Review the confirmation dialog, try submitting with the box unticked to see the error, then tick it and submit.
5. Open the **Application Tracker** to see the new application in Applied, then use the demo button to advance it.
6. Open **Preparation Hub**, log a DSA problem and watch the progress bar and the Dashboard readiness score change.

Real in the prototype	Mocked or simplified
Eligibility rules and their pass or fail logic	Companies, deadlines and notifications
Filters, search and sort	Stage changes (advanced by a demo button)
Form validation and file-type checks	Resume upload (the file is not stored)
Keyboard operation and ARIA labels	No accounts, no saved data between visits

10. Conclusion

CareerFlow brings discovery, eligibility, preparation, application and tracking into one student-centred portal. The design follows the ISO 9241-210 cycle: the context of use and tasks were analysed, requirements were prioritised, and solutions were developed from information architecture and flows through sketches and wireframes to an interactive prototype. HCI principles from the course (visibility of status, recognition over recall, Fitts's law, Gestalt grouping, error prevention, consistency, universal design and feedback) were applied and traced to specific parts of the interface.

The evaluation so far is an expert review, which found ten issues. Two are rated 3 and concern saved jobs and keyboard focus in dialogs. User testing has not been done, and the personas are assumptions. The next steps are therefore to validate the personas with students, apply the improvements in section 8, run the usability test in section 7.2, and then extend the design to placement officers.

References

- ISO 9241-210:2019, Ergonomics of human-system interaction: Human-centred design for interactive systems.
- ISO 9241-11:2018, Usability: Definitions and concepts.
- Nielsen, J. (1994). Ten usability heuristics for user interface design.
- Shneiderman's Eight Golden Rules and Norman's Seven Principles, as covered in the course (Module 3).
- Web Content Accessibility Guidelines (WCAG) 2.1, W3C.
- Course slides: HCI Modules 1 to 5 (task analysis, design rules, universal design, evaluation techniques).